SQL(Advanced)

Assignment

Problem Statement:

You have been assigned to work as a Database Developer for an online shopping platform. The platform manages customer, product, and order information in a relational database and has experienced rapid growth in both data volume and user activity.

As part of the database modernization initiative, your responsibility is to develop advanced SQL solutions that support business reporting, sales analysis, database automation, transaction management, and query optimization. The solution should efficiently retrieve and analyze data, improve database performance, ensure data consistency, and automate common business operations.

The objective of this assignment is to demonstrate your understanding of advanced SQL concepts, including advanced joins, subqueries, set operations, window functions, Common Table Expressions (CTEs), views, indexes, stored procedures, functions, transactions, triggers, and performance tuning.

Create the required database tables, populate them with sample data, and complete the tasks listed below.

Database Schema

Customers

- customer_id
- customer_name
- city
- email

Products

- product_id
- product_name
- category
- price

Orders

- order_id
- customer_id
- product_id
- quantity
- order_date

Populate the tables with:

- At least 10 customers
- At least 10 products
- At least 20 orders

Tasks

Section 1: Advanced Joins

- Task 1: Display each customer along with the products they ordered.
- Task 2: Display all possible combinations of customers and products using a CROSS JOIN.
- Task 3: Display all customers and their orders, including customers who have not placed any orders.

Section 2: Subqueries

- Task 4: Find products priced above the average product price.
- Task 5: Display customers whose total purchase amount is greater than the average customer purchase amount.
- Task 6: Find the highest-priced product in each category using a correlated subquery.

Section 3: Set Operations

- Task 7: Combine products from two different categories using UNION.
- Task 8: Write MySQL alternatives for INTERSECT and EXCEPT using INNER JOIN and LEFT JOIN.

Section 4: Window Functions

- Task 9: Rank products by price within each category.
- Task 10: Display the previous product price (LAG) and a running total of product prices using window functions.

Section 5: Common Table Expressions (CTEs)

- Task 11: Create a CTE to display customers whose total purchase amount is above the average.
- Task 12: Create a recursive CTE to generate a sequence of numbers from 1 to 10.

Section 6: Views

- Task 13: Create a view that displays products priced above ₹1,000 and retrieve data from the view.

Section 7: Indexes and Query Performance

- Task 14: Create an index on the category column of the Products table.
- Task 15: Use EXPLAIN to analyze a query that filters products by category.

Section 8: Stored Procedures and Functions

- Task 16: Create a stored procedure to retrieve products by category.
- Task 17: Create a function to calculate the total price of an order (price × quantity).

Section 9: Transactions

- Task 18: Write a transaction that updates a product's price and demonstrate the use of COMMIT and ROLLBACK.

Section 10: SQL Triggers

- Task 19: Create a trigger that automatically records new orders in an Order_Audit table whenever a new order is inserted.

Section 11: Performance Tuning

- Task 20: Optimize the queries by applying SQL performance tuning best practices:
  eg:

```
SELECT DISTINCT *
FROM Products
WHERE YEAR(created_date) = 2024
  AND category = 'Electronics';
```

  Apply the following improvements:
  1. Avoid SELECT *.
  2. Remove unnecessary DISTINCT.
  3. Avoid functions on indexed columns.
  4. Use an index-friendly query.

Deliverables:

Submit the following as part of your assignment:

A single .sql file containing:

- CREATE TABLE statements
- Sample INSERT statements
- Solutions for all assignment tasks
- Database objects such as Views, Indexes, Stored Procedures, Functions, and Triggers
- Organize the SQL script section-wise (Advanced Joins, Subqueries, Set Operations, Window Functions, CTEs, Views, Indexes & Performance, Stored Procedures & Functions, Transactions, Triggers, and Performance Tuning) with appropriate task numbering and comments (for example, -- Section 1: Advanced Joins, -- Task 1, -- Task 2, etc.).

Upload the .sql file to a GitHub repository.
Share the following:

- GitHub Repository link
- Pull Request (PR) link
- Branch name used for implementation

A README.md containing:

- Assignment overview
- Database schema
- Steps to execute the SQL script
- Summary of the implemented advanced SQL concepts

Screenshots showing the successful execution of:

- Advanced Join query
- Subquery
- Window Function
- CTE
- View
- Stored Procedure or Function
- Trigger execution
- Transaction (COMMIT / ROLLBACK)
- EXPLAIN query output

Share daily status updates through email during the assignment.

[End]